

Centre Number

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Student Number

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**2021
/2022**

Year 12

Mathematics Extension 2

Task 1
2/12/21

Q	Marks
1	/14
2	/15
3	/15
Total	/44

General
Instructions:

- Reading time – 60 minutes
- Working time – 5 minutes
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations
- Answer all questions in the booklets provided

Total Marks:
44

This question paper must not be removed from the examination room.

This assessment task constitutes 25% of the course.

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In Questions 1 – 3, your responses should include relevant mathematical reasoning and/or calculations.

Question 1: Answer Question 1 in a new booklet. (14 Marks)

- (a) Given the complex number $z = 3 - 3i$ and $w = -\sqrt{3} + i$:
- (i) Find $\arg(z)$ 1
- (ii) Find zw in the form $a + ib$ 1
- (iii) Find $\frac{z}{w}$ in the form $a + ib$ 2
- (iv) Find $\left| \frac{z^2}{w^4} \right|$ 2
- (v) Find zw in Mod-Arg form. 2
- (vi) Hence, find the exact value of $\tan \frac{7\pi}{12}$ 1
- (vii) Sketch z , w and zw on the Argand plane supplied as a separate answer sheet. 2
- (b) Find real numbers a and b such that $z = 3 + i$ satisfies $z + \frac{a}{z} = b$. 3

Question 2: Answer Question 2 in a new booklet. (15 Marks)

- (a) (i) Find $\sqrt{6 + 8i}$ 3
- (ii) Hence, or otherwise, solve $z^2 + \sqrt{2}iz + (-2 - 2i) = 0$, $z \in \mathbb{C}$. 1
- (b) Give the linear factorisation of $4x^3 + 2x^2 - 14x + 3$, $x \in \mathbb{R}$, given that it has one rational zero. 3
- (c) The complex numbers $z_1 = 2 - 5i$, $z_2 = 1 + 3i$ and z_3 are represented in the Argand plane by the points P , Q and R respectively. Find all possible values of z_3 in the form $a + ib$, such that $OPQR$ is a parallelogram. 3
- (d) (i) Sketch the region in the complex plane that satisfies the following three inequalities: 3
- $$|z - 2i| \leq \sqrt{2}$$
- $$\operatorname{Im}((1 + i)z) < 2$$
- (ii) Hence, or otherwise, find the range of $\arg(z)$. 2

Question 3: Answer Question 3 in a new booklet. (15 Marks)

(a) For which real values of c is $z - ci$ a factor of $P(z) = z^4 - z^3 + 9z^2 - 4z + 20$? **3**

(b) Consider the non-zero complex numbers z and w . **3**

Given that $\sqrt{wz}$ is real, show that $\frac{z}{|z|} = \frac{|w|}{w}$.

(c) (i) Expand $(x - 1)(1 + x + x^2 + \cdots + x^n)$ **1**

(ii) Hence, or otherwise, show that $1 + x + x^2 + \cdots + x^n$ has no multiple roots, $n \geq 0$. **3**

(d) Consider the complex number z , with a locus that describes the horizontal line:

$$\begin{aligned}x &= a \\ a &\in \mathbb{R}\end{aligned}$$

(i) Find the locus of $w = z^2 + z$ and describe its shape in geometric terms. **3**

(ii) Describe the locus of w when $a = -\frac{1}{2}$. **2**

End of Examination

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Killara
HIGH SCHOOL

MATHEMATICS DEPARTMENT

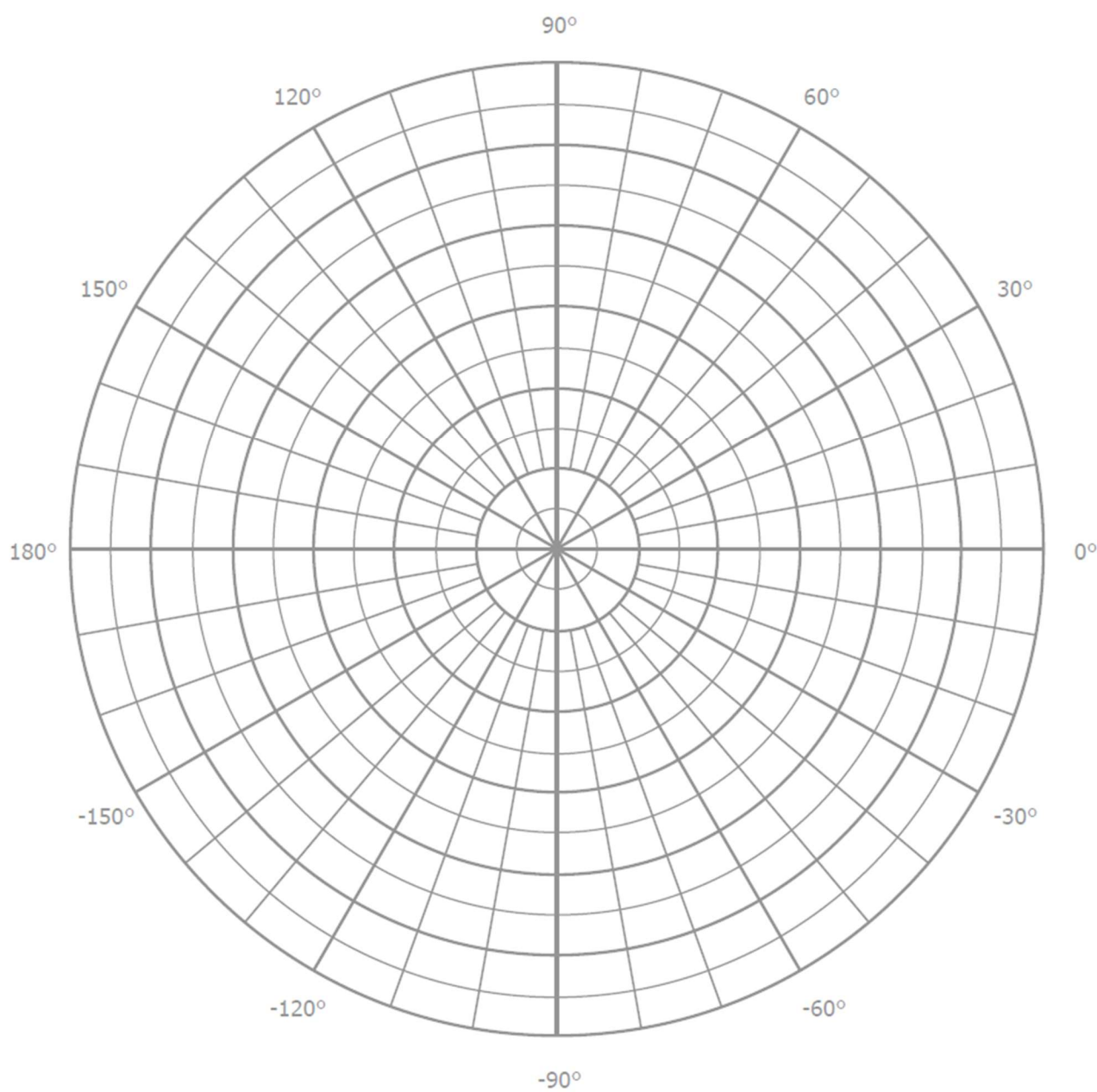
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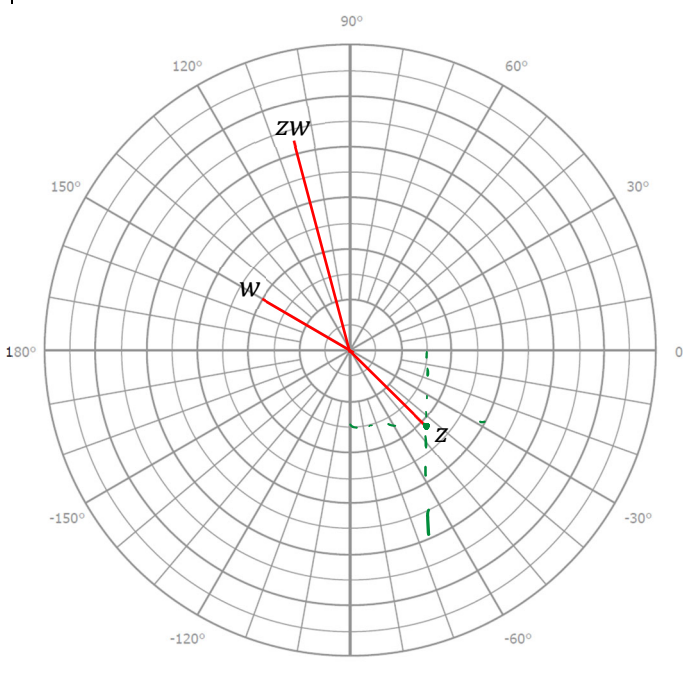
Argand plane for Question 1 Part a (vii)



2021/2022 Ext 2 Task 1 – Solutions and Feedback

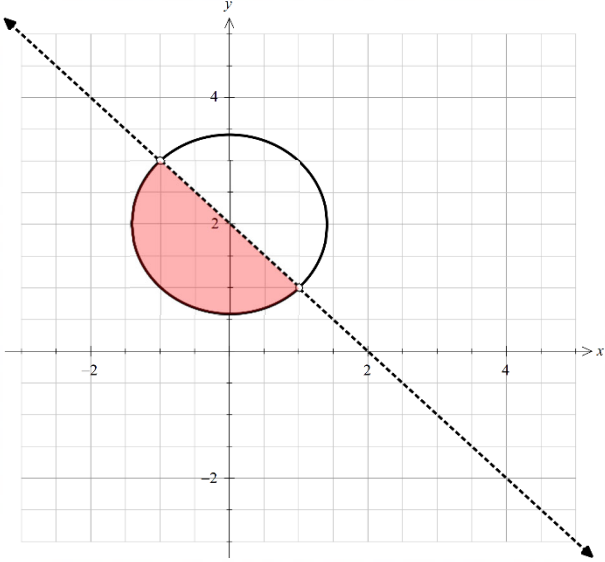
Question 1

Q	Solutions	Marking Scheme	Feedback
(ai)	$\sin(\arg(z)) < 0$ $\cos(\arg(z)) > 0$ $\therefore -\frac{\pi}{2} < \arg(z) < 0$ $\arg(z) = \tan^{-1} \frac{-3}{3}$ $= -\frac{\pi}{4}$	1 mark for solution with working	All good
(ii)	$zw = (3 - 3i)(-\sqrt{3} + i)$ $= -3\sqrt{3} + 3 + (3\sqrt{3} + 3)i$	1 mark for solution with working	All good
(iii)	$\frac{z}{\bar{w}} = \frac{zw}{\bar{w}zw}$ $= \frac{zw}{ w ^2}$ $= \frac{-3\sqrt{3} + 3 + (3\sqrt{3} + 3)i}{(-\sqrt{3})^2 + 1^2}$ $= \frac{-3\sqrt{3} + 3 + (3\sqrt{3} + 3)i}{4}$ $= \frac{-3\sqrt{3} + 3}{4} + \frac{3\sqrt{3} + 3}{4}i$	1 mark for multiplying by $\frac{w}{w}$ 1 mark for solution with working (<i>a</i> and <i>b</i> must be separated fractions)	Mostly well done
(iv)	$\left \frac{z^2}{w^4} \right = \frac{ z^2 }{ w^4 }$ $= \frac{ z ^2}{(w ^2)^2}$ $= \frac{3^2 + (-3)^2}{4^2}$ $= \frac{18}{16}$ $= \frac{9}{8}$	1 mark for expressing in terms of z and w 1 mark for solution with working	Using de Moivre's theorem was ok. Some made this harder than it needed to be.
(v)	$ z = \sqrt{18}$ $= 3\sqrt{2}$ $\arg(z) = -\frac{\pi}{4}$ $z = 3\sqrt{2} \left(\cos\left(-\frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{4}\right) \right)$ $ w = \sqrt{4}$ $= 2$ $\arg(w) = \tan^{-1} \frac{1}{-\sqrt{3}} \text{ (in QII)}$ $= \frac{5\pi}{6}$ $w = 2 \left(\cos\left(\frac{5\pi}{6}\right) + i \sin\left(\frac{5\pi}{6}\right) \right)$ $wz = 6\sqrt{2} \left(\cos\left(\frac{7\pi}{12}\right) + i \sin\left(\frac{7\pi}{12}\right) \right)$	1 mark for expressing <i>z</i> and <i>w</i> in Mod-Arg form 1 mark for solution with working	Some did not show full working. Avoid Cis for calculations and results
(vi)	$\tan\left(\frac{7\pi}{12}\right) = \frac{3\sqrt{3} + 3}{-3\sqrt{3} + 3}$ $= \frac{\sqrt{3} + 1}{-\sqrt{3} + 1}$ $= -2 - \sqrt{3}$	1 mark for solution with working	Mostly well done

(vii)	Note: $6\sqrt{2} \approx 8.5$ 	1 mark for any one correct 2 marks for all three correct	Poorly done. Often no use of scale.
(b)	$z + \frac{a}{z} = b$ $(3 + i) + \frac{a}{3 + i} = b$ $(3 + i)^2 + a = b(3 + i)$ $9 - 1 + a + 6i = 3b + bi$ $(8 + a) + 6i = 3b + bi$ Hence, $b = 6$ equating imaginary components $8 + a = 3b$ equating real components $a = 10$ $\therefore a = 10, b = 6$	1 mark for substituting $z = 3 + i$ 1 mark for equating two complex numbers in $a + ib$ form 1 mark for equating components and finding values of a and b	Many tried to solve for z before substituting making it harder.

Question 2

Q	Solutions	Marking Scheme	Feedback
(ai)	Let $\sqrt{6+8i} = x + yi$ such that $6 + 8i = (x + yi)^2$ $= x^2 - y^2 + 2xyi$ $x^2 - y^2 = 6$ (1) $xy = 4$ $y = \frac{4}{x}$ (2) $x^2 + y^2 = \sqrt{6^2 + 8^2}$ $= 10$ (3) (1)+(2) $2x^2 = 16$ $x = \pm 2\sqrt{2}$ When $x = 2\sqrt{2}$ Sub in (2) $y = \frac{2}{\sqrt{2}}$ $= \sqrt{2}$ When $x = -2\sqrt{2}$ Sub in (2) $y = \frac{2}{-\sqrt{2}}$ $= -\sqrt{2}$ $\therefore \sqrt{6+8i} = \pm(2\sqrt{2} + \sqrt{2}i)$	1 mark for composing any two of the equations (1), (2) or (3) 1 mark for finding solutions for x or y 1 mark for stating both possibilities as $\pm(2\sqrt{2} + \sqrt{2}i)$	Mostly well done
(ii)	$z^2 + \sqrt{2}iz + (-2 - 2i) = 0$ $z = -\frac{\sqrt{2}i \pm (2\sqrt{2} + \sqrt{2}i)}{2}$ $= -\frac{2\sqrt{2} + 2\sqrt{2}i}{2} \text{ or } \frac{2\sqrt{2}}{2}$ $= -\sqrt{2} - \sqrt{2}i \text{ or } \sqrt{2}$	1 mark for solutions with appropriate working	Mostly well done
(b)	Let $P(x) = 4x^3 + 2x^2 - 14x + 3$ By rational root theorem, possible zeroes exist for $x = \pm 3, \pm 1, \pm \frac{3}{2}, \pm \frac{1}{2}, \pm \frac{3}{4}, \pm \frac{1}{4}$ By substitution, $P\left(\frac{3}{2}\right) = 0$ Which gives the factor $2x - 3$. $P(x) = (2x - 3)(ax^2 + bx + c)$ $a = 2$ (coefficients of x^4) $c = -1$ (constant terms) $-3b + 2c = -14$ (coefficients of x) $b = 4$ $\therefore P(x) = (2x - 3)(2x^2 + 4x - 1)$ Now $2x^2 + 4x - 1 = 2(x^2 + 2x) - 1$ $= 2(x^2 + 2x + 1) - 3$ $= (\sqrt{2}(x + 1))^2 - (\sqrt{3})^2$ $= (\sqrt{2}(x + 1) + \sqrt{3})(\sqrt{2}(x + 1) - \sqrt{3})$ $= 2(x + 1 + \sqrt{6})(x + 1 - \sqrt{6})$ $P(x) = 2(2x - 3)(x + 1 + \sqrt{6})(x + 1 - \sqrt{6})$ $P(x) = 4(x - \frac{3}{2})(x + 1 + \sqrt{6})(x + 1 - \sqrt{6})$	1 mark – finds rational zero 1 mark – factorises $P(x)$ into linear and quadratic factors 1 mark – linear factorisation (monic linear factors)	Many were lost on this question, they could not find the rational root $x = \frac{3}{2}$ As well, some did not go all the way to linear factors.

(c)	For a parallelogram, $z_3 = z_1 + z_2$ or $z_3 = z_1 - z_2$ or $z_3 = z_2 - z_1$ <u>Case 1</u> $z_3 = z_1 + z_2$ $= 3 - 2i$ <u>Case 2</u> $z_3 = z_1 - z_2$ $= 1 - 8i$ <u>Case 3</u> $z_3 = z_2 - z_1$ $= -1 + 8i$	1 mark for each solution with appropriate working	Many missed at least one solution
(di)		1 mark for circle 1 mark for linear relationship 1 mark for area with excluded values	Many missed the open circles on the circle.
(ii)	$\frac{\pi}{4} < \arg(z) \leq \frac{3\pi}{4}$	1 mark for maximum 1 mark for minimum (with correct inequality)	Mostly ok when attempted.

Question 3

Q	Solutions	Marking Scheme	Feedback
(a)	$x - ci$ is a factor implies: $P(ci) = 0$ $(ci)^4 - (ci)^3 + 9(ci)^2 - 4(ci) + 20 = 0$ $c^4 + c^3i - 9c^2 - 4ci + 20 = 0$ $(c^4 - 9c^2 + 20) + (c^3 - 4c)i = 0 + 0i$ $c^4 - 9c^2 + 20 = 0 \text{ (1) and } c^3 - 4c = 0 \text{ (2)}$ From (2) $c(c^2 - 4) = 0$ $c(c + 2)(c - 2) = 0$ $c = 0, \pm 2$ From (1) $c^4 - 9c^2 + 20 = 0$ $(c^2 - 4)(c^2 - 5) = 0$ $c = \pm 2, \pm \sqrt{5}$ To satisfy both (1) and (2) $c = \pm 2$	1 mark for substituting ci into $P(x)$ 1 mark for equating real and imaginary components 1 mark for solutions with appropriate working	- Mostly well done - Some tried conjugate root theorem, although factor theorem was much more relevant here - Testing for integer solutions was not a valid method as it assumed an integer solution, when this was not given in the question. - Some students equated real components, and unfortunately ignored imaginary components
(b)	$\sqrt{wz} \in \mathbb{R}, w \neq 0, z \neq 0$ implies $wz \in \mathbb{R}^+$ Hence, $\arg(z) + \arg(w) = 0$ $\arg(z) = -\arg(w)$ $\arg(z) = \arg(\bar{w})$ Now $\frac{z}{ z } = \cos(\arg(z)) + i \sin(\arg(z))$ $= \cos(\arg(\bar{w})) + i \sin(\arg(\bar{w}))$ $= \frac{\bar{w}}{ \bar{w} }$ $= \frac{\bar{w}}{ w }$ $= \frac{\bar{w} w }{ w ^2}$ $= \frac{\bar{w} w }{w\bar{w}}$ $= \frac{\bar{w}}{w}$	1 mark for $wz \in \mathbb{R}^+$ 1 mark for $\arg(z) = \arg(\bar{w})$ 1 mark for converting between $\frac{ w }{w}$ and $\frac{\bar{w}}{ w }$ Or equivalent	- This was a difficult conceptual question. - Many went straight into the algebra without considering the consequences of the given condition $\sqrt{wz} \in \mathbb{R}$ - Some assumed that $w = \bar{z}$, which was not the case, as the moduli were different. However, dividing by the moduli would have led to the result (c.f. unit vectors)
(b)	Alternatively, Let $\alpha = \sqrt{wz}, \alpha \in \mathbb{R}$ $\alpha^2 = wz$ $= \sqrt{wz}(\sqrt{wz})$ $= \sqrt{wz\bar{w}\bar{z}}$ $= \sqrt{w\bar{w}z\bar{z}}$ $= \sqrt{ w ^2 z ^2}$ $= w z $ $\therefore wz = w z $ Hence, $\frac{z}{ z } = \frac{ w }{w}$	1 mark for considering $\sqrt{wz}(\sqrt{wz})$ 1 mark for manipulating to $\sqrt{w\bar{w}z\bar{z}}$ 1 mark for considering $wz = w z$ and hence justifying the given statement	- Noting that $wz \in \mathbb{R}^+$ could lead to $wz = wz$, which could have led quickly to the result

(ci)	$(x-1)(1+x+x^2+\dots+x^n)$ $= x+x^2+x^3+\dots+x^n+x^{n+1}$ $-1-x-x^2-x^3-\dots-x^n$ $= x^{n+1}-1$	1 mark	Mostly well done. This would be a good place to using two rows for your working of the expansion
(ii)	Roots of $1+x+x^2+\dots+x^n$ are also roots of $x^{n+1}-1$ Let $P(x) = x^{n+1}-1$ $P'(x) = (n+1)x^n$ $P'(x) = 0$ only when $x = 0$ (for $n \geq 0$) But $P(0) = 0-1$ $= -1$ $\therefore x = 0$ is not a root of $P(x)$ and $P(x)$ has no multiple roots and thus $1+x+x^2+\dots+x^n$ has no multiple roots.	1 mark for comparing roots of $1+x+x^2+\dots+x^n$ and roots of $x^{n+1}-1$ 1 mark for taking derivative and letting $P(x) = 0$ 1 mark for stating contradiction and logically proving result (Or equivalent method).	- Many missed the connection between part (i) and part (ii). Explaining this connection was required for the first mark and not many achieved this. - Students should define $P(x)$. Many took the derivative without stating what $P(x)$ was. Were you using $P(x) = x^{n+1}-1$ - OR $P(x) = 1+x+x^2+\dots+x^n$ - Dividing $\frac{x^{n+1}-1}{x-1}$ let to a discontinuity in the function, which changed the nature of the question. - A few students started finding roots of n, although the polynomial is in x.
(di)	Let $z = a + iy$ $w = z^2 + z$ $= (a + iy)^2 + (a + iy)$ $= a^2 + a - y^2 + (2ay + y)i$ Consider $w = X + iY$, where $X = a^2 + a - y^2$ (1) and $Y = 2ay + y$ (2) $y = \frac{Y}{2a+1}, a \neq -\frac{1}{2} \text{ from (2)}$ Sub in (1) $X = a^2 + a - \left(\frac{Y}{2a+1}\right)^2$ $= a^2 + a - \frac{Y^2}{(2a+1)^2}$ $Y^2 = -(2a+1)^2(X - (a^2 + a))$ This describes a concave left parabola, for $a \neq -\frac{1}{2}$. (vertex at $(a^2 + a, 0)$)	1 mark for establishing $z = a + iy$ 1 mark for relating X to Y 1 mark for describing shape as parabola, based on sufficient working/justification	- Many achieved the first mark by establishing $z = x + iy$ and substituting $x = a$ - Not many took the next step of observing y as a parametric - Stated the shape as a parabola without justification did not achieve the mark
(ii)	When $a = -\frac{1}{2}$ $X = -\frac{1}{4} - y^2 \text{ from (1)}$ $Y = 0 \text{ from (2)}$ This describes the X-axis for $X \leq -\frac{1}{4}$.	1 mark for shape 1 mark for condition $X \leq -\frac{1}{4}$	- Disappointingly, many did not attempt this question. However, this could have been achieved relatively easily, after the straightforward algebraic step of the first mark of part (i).